GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 3

By- SIDDHARTH SABHARWAL SIR

Recap of Previous Lecture

date 12/9/2024

Arguments.

- Topic
- Topic
- Topic
- Topic
- Topic

Turn on Slide map

Topics to be Covered

- Topic
- Topic
- Topic
- Topic
- Topic

Questions

Today is your opportunity to build the tomorrow you want.
—KEN POIROT
RD

[MCQ]

Which of the following implication is/are valid

or

which of the following is/are a tautology

done

A $(P \wedge Q) \rightarrow (P \leftrightarrow Q)$

B $(P \leftrightarrow Q) \rightarrow (P \rightarrow Q)$

C $(P \leftrightarrow Q) \rightarrow (P \rightarrow \sim Q)$

D $Q \rightarrow (P \rightarrow Q)$

[MCQ]

Which of the following is/are false

A. $a \to (b \vee c) \equiv (a \wedge \sim b) \to c$

B. $(P \to Q) \wedge (R \to Q) \equiv (P \vee R) \to Q$

C. $(P \to Q) \vee (R \to Q) \equiv (p \wedge R) \to Q$

done

1 0

$a \to (b \vee c)$

$b=0$
$c=0$

abc
100

$(a \wedge \sim b) \to c$

1 0

$a=1$
$b=0$

$c=0$

false 100

[MCQ]

Which of the following is/are false

A. $a \to (b \vee c) \equiv (a \wedge \sim b) \to c$ — True

B. $(P \to Q) \wedge (R \to Q) \equiv (P \vee R) \to Q$ — True

C. $(P \to Q) \vee (R \to Q) \equiv (p \wedge R) \to Q$ — True

3. $(P \to Q) \vee (R \to Q) \Rightarrow$ False only @5

Zero @ 4,5 — Zero @ 1,5

$(P \wedge R) \to Q \Rightarrow$ False @5

$(P \wedge R)$: 1 — Q: 0

PQR: 101

2. $(P \to Q) \wedge (R \to Q) \Rightarrow$ False @ (1, 4, 5)

$(P \to Q) \Downarrow$

P	Q	R
1	0	0
1	0	1

(4,5)

$(R \to Q)$

P	Q	R
1	0	1
0	0	1

1,5

Same

$(P \vee R) \to Q \Rightarrow$ False @ (1,4,5)

$(P \vee R)$: 1 — Q: 0

P	Q	R
0	0	1
1	0	0
1	0	1

Argument ⇒

Argument Can be written as

1. $(P_1, P_2, P_3 \ldots P_N) \underbrace{\longmapsto}_{\text{yields}} Q$

$(P_1, P_2, P_3 \ldots P_N)$ yields Q

2. $(P_1, P_2, \ldots P_N)$ logically implies Q.

[MCQ]

Which of the following arguments is/are valid

(A) Conclusion R follows ~~from~~ from the premises $\{P \to (Q \to R), P \wedge Q\}$

$\underbrace{P \to (Q \to R)}_{1 \quad 0}$, $P \wedge Q$ (0, 2, 4, 1, 3, 5)

1 0 false false

P	Q	R
1	1	0

6

P	Q	R
0	0	0
0	1	0
1	0	0
0	0	1
0	1	1
1	0	1

~~0, 1, 2, 3, 4, 5, 6~~ 7

P	Q	R
1	1	1

So valid argument

[MCQ]

Which of the following arguments is/are valid

$P \to (R \to S)$	$\sim R \to \sim P$	P false
1 0 1 0	1 0	PRS
PRS	PRS	0 0 0 (0 1 2 3)
1 1 0 ← False(6)	1 0 0 (4,5)	0 0 1
	1 0 1	0 1 0
		0 1 1

~~0, 1, 2, 3~~
~~4, 5, 6~~, 7

B $\{P \to (R \to S), \sim R \to \sim P, P\}$ logically implies 'S'

Critical Row

7 PRS

1 1 1

[MCQ]

Which of the following arguments is/are valid

$P \vee Q$, $\sim P$

P	Q
0	0

False @ zero

P	Q
1	0
1	1

False @ 2, 3

~~(a)~~, 1, ~~(c)~~, ~~(d)~~

(c) $(P \wedge Q)$ follows from the premises $\{(P \vee Q), \sim P\}$

P	Q
0	1

← Critical Row

$P \wedge Q$: 0

Invalid

[MCQ]

Which of the following arguments is/are valid

- Any sequence : allowed
- Same sequence to be used for all premise

$(\sim T \to \sim R)$

False 1 0

T=0 R=1

W	R	S	T
0	1	0	0
0	1	1	0
1	1	0	0
1	1	1	0

(4, 6, 12, 14)

$\sim S$

False S=1

W	R	S	T
0	0	1	0
0	0	1	1
0	1	1	0
0	1	1	1
1	0	1	0
1	0	1	1
1	1	1	0
1	1	1	1

(2, 3, 6, 7, 10, 11, 14, 15)

$T \to W$

False 1 0

W	R	S	T
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1

(1, 3, 5, 7)

$R \vee S$

False R=0, S=0

W	R	S	T
0	0	0	0
0	0	0	1
1	0	0	0
1	0	0	1

(0, 1, 8, 9)

(D) $\{\sim T \to \sim R, \sim S, T \to W, R \vee S\} \mapsto W$

~~0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15~~

13 is Critical Row

W	R	S	T
1	1	0	1

w=1

[NAT]

Check whether the following argument is valid or invalid

$\{P \to Q, \sim(P \wedge Q)\}$ yields $\sim P$

$P \to Q$, $\sim(P \wedge Q)$

False $P=1$, $Q=0$

$10 \Rightarrow 2$

P Q	$\sim(P \wedge Q)$
0 0	1
0 1	1
1 0	1
1 1	0

False @ $PQ = 1.1$ $\Rightarrow 3$

$PQ = \boxed{0,1}, \cancel{1,1}, \cancel{1,0}$

Concl. true.

	P Q	$\sim P$
Critical Row ✓	0 0	1
Critical Row ✓	0 1	1

$$\underbrace{(\underbrace{A}_{\text{antecedant}} \rightarrow \underbrace{B}_{\text{Consequent}})}$$

$$\underbrace{\left(\overset{1}{(P \wedge Q)} \to \overset{0}{R}\right)}_{\text{False}} \to \underbrace{\left(\overset{1}{(P \wedge Q)} \to \overset{0}{(Q \wedge R)}\right)}_{\text{False}}$$

P	Q
0	0
0	1
1	0

P	Q	R
1	1	0

equivalent

P	Q	R
1	1	0

PQR	A: $(P \wedge Q) \to R$	B: $(P \wedge Q) \to (Q \wedge R)$	$A \to B$
0	T	T	T
1	T	T	T
2	T	T	T
3	T	T	T
4	T	T	T
5	T	T	T
6	F	F	T
7	T	T	T

Q1 Choose the correct choice(S) regarding the following proportional logic assertion S:

S: $((P \wedge Q) \to R) \to ((P \wedge Q) \to (Q \wedge R))$

(A) S is neither a tautology nor a contradiction.

(B) S is a tautology.

(C) S is a contradiction.

(D) The antecedent of S is logically equivalent to the consequent of S.

Q2 Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by

(A) $(\neg p \wedge r) \wedge (\neg r \to (p \wedge q))$

(B) $(\neg p \wedge r) \wedge ((p \wedge q) \to \neg r)$

(C) $(\neg p \wedge r) \vee ((p \wedge q) \to \neg r)$

(D) $(\neg p \wedge r) \vee (r \to (p \wedge q))$

$$\overset{1}{(}\overset{0}{(P \wedge Q) \to R)} \to \overset{11}{(}\underset{1}{(P \wedge Q)} \to \overset{1\ 0}{(}\underset{0}{(Q \to R)})$$

False

P	Q	R
1	1	0

@6

False

P	Q	R
1	1	0

False @6 only

Some

Q2. P: It is Raining
Q: It is Cold
R: It is pleasant

$(\neg P \wedge R) \wedge (\neg R$

Q1 Choose the correct choice(S) regarding the following proportional logic assertion S:

S: $((P \wedge Q) \to R) \to ((P \wedge Q) \to (Q \to R))$

(A) S is neither a tautology nor a contradiction.

(B) S is a tautology.

(C) S is a contradiction.

(D) The antecedent of S is logically equivalent to the consequent of S.

Q2 Let p, q, r denote the statements "It is raining", "It is cold", and "It is pleasant", respectively. Then the statement "It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold" is represented by

(A) $(\neg p \wedge r) \wedge (\neg r \to (p \wedge q))$

(B) $(\neg p \wedge r) \wedge ((p \wedge q) \to \neg r)$

(C) $(\neg p \wedge r) \vee ((p \wedge q) \to \neg r)$

(D) $(\neg p \wedge r) \vee (r \to (p \wedge q))$

(A only if B ⇒ A → B
A if B ⇒ B → A)

1) (A only if B)

→ ~~English~~ ~~B → A~~

⇒ (A → B)

~~B → A~~

- A Tabhi aega Jab B hoga
- nahi to nahi aega

2) (A if B) ⇒ (B → A)

↳ So A is 1 when B=1 but here A can be 1 even if B=0.

A	B	A→B
0	0	1
0	1	1
1	0	0
1	1	1

A	B	B→A
0	0	1
0	1	0
1	0	1
1	1	1

$\neg R = 0$ $\neg S = 0$

$\neg((p \to q) \wedge (\neg r \vee \neg s))$

False

P Q R S
1 0 0 0
1 0 0 1
1 0 1 0
1 0 1 1

8,9,10,11

P Q R S
0 0 1 1
0 1 1 1
1 0 1 1
1 1 1 1

3,7,11,15

False → (3,7,8,9,10,11,15)

Overall true 3,7,8,9,10,11,15

Q5 Let p, q, r, s represent the following propositions.

p : x ∈ {8, 9, 10, 11, 12}

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer x ≥ 2 which satisfies ¬((p ⇒ q) ∧ (¬r ∨ ¬s)) is 11 _____.

P Q R S
0 0 1 1
0 1 1 1
1 0 0 0
1 0 0 1
1 0 1 0
1 0 1 1
1 1 1 1

Q6 Which one of the following Boolean expressions is NOT a tautology?

(A) ((a → b) ∧ (b → c)) → (a → c)

(B) (a ↔ c) → (~b → (a ∧ c))

(C) (a ∧ b ∧ c) → (c ∨ a)

(D) a → (b → a)

a) $\overbrace{\left[\overset{10}{(a \to b)} \wedge \overset{10}{(b \to c)}\right]}^{P} \to \overbrace{\overset{10}{[a \to c]}}^{Q}$

abc	abc	abc
100	110	100
101	010	110
4,5	6,2	(4,6)
False	(2,4,5,6)	False

abc	P	Q	P→Q
0	T	T	T
1	T	T	T
2	F	T	T
3	T	T	T
4	F	F	T
5	F	T	T
6	F	F	T
7	T	T	T

Q5 Let p, q, r, s represent the following propositions.

p : x ∈ {8, 9, 10, 11, 12}

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer x ≥ 2 which satisfies ¬((p ⇒ q) ∧ (¬r ∨ ¬s)) is 11 _____.

P	Q	R	S
0	0	1	1
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	1	1

Q6 Which one of the following Boolean expressions is NOT a tautology?

(A) ((a → b) ∧ (b → c)) → (a → c)

(B) (a ↔ c) → (~b → (a ∧ c))

(C) (a ∧ b ∧ c) → (c ∨ a)

(D) a → (b → a)

P

1 0

0 1

$a \leftrightarrow c$

a	b	c
0	0	1
0	1	1
1	1	0
1	0	0

False (1,3,6,4)

Q

1 0

$\sim b \rightarrow (a \wedge c)$

$b=0$

False

a	c
0	1
0	0
1	0

a	b	c
0	0	0
0	0	1
1	0	0

False

0,1,4

a	b	c	P	Q
0			T	F → F
1			F	F
2			T	T
3			F	T
4			F	F
5			T	F
6			F	T
7			T	T

Q5 Let p, q, r, s represent the following propositions.

p : x ∈ {8, 9, 10, 11, 12}

q : x is a composite number

r : x is a perfect square

s : x is a prime number

The integer x ≥ 2 which satisfies ¬((p ⇒ q) ∧ (¬r ∨ ¬s)) is _____.

P	Q	R	S
0	0	1	1
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	1	1

Q6 Which one of the following Boolean expressions is NOT a tautology?

(A) $((a \rightarrow b) \wedge (b \rightarrow c)) \rightarrow (a \rightarrow c)$

(B) $(a \leftrightarrow c) \rightarrow (\sim b \rightarrow (a \wedge c))$

(C) $(a \wedge b \wedge c) \rightarrow (c \vee a)$

(D) $a \rightarrow (b \rightarrow a)$

THANK - YOU